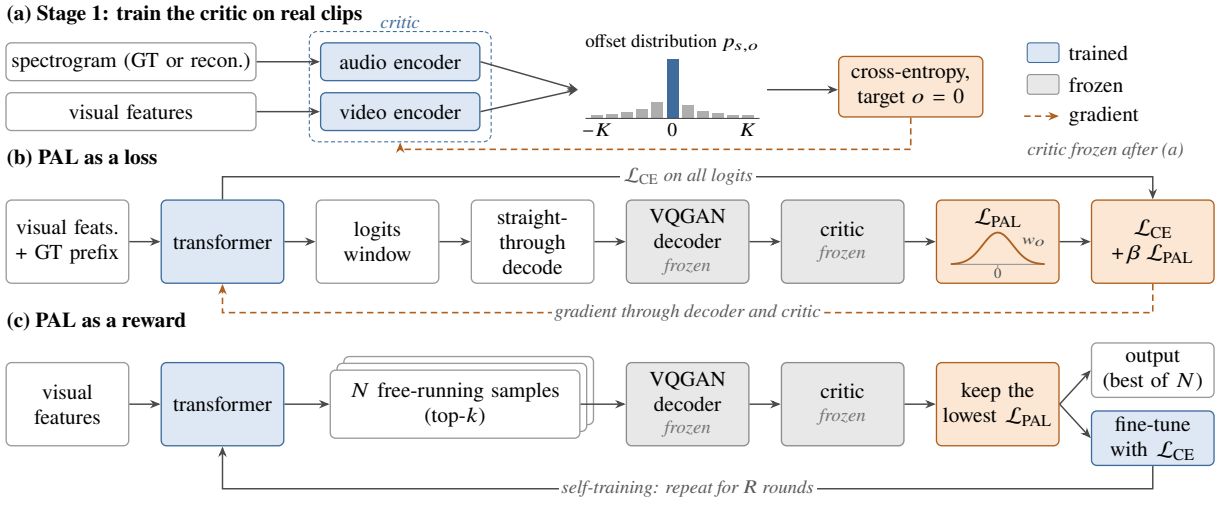
(a) Stage 1: train the critic on real clips
critic
spectrogram (GT or recon.)
visual features
audio encoder
video encoder
offset distribution $p_{s,o}$
$-K$
0
K
cross-entropy, target $o = 0$
trained
frozen
gradient
critic frozen after (a)
(b) PAL as a loss
$\mathcal{L}_{\text{CE}}$ on all logits
visual feats. + GT prefix
transformer
logits window
straight-through decode
VQGAN decoder frozen
critic frozen
$\mathcal{L}_{\text{PAL}}$
w_o
0
$\mathcal{L}_{\text{CE}} + \beta\,\mathcal{L}_{\text{PAL}}$
gradient through decoder and critic
(c) PAL as a reward
visual features
transformer
N free-running samples (top-k)
VQGAN decoder frozen
critic frozen
keep the lowest $\mathcal{L}_{\text{PAL}}$
output (best of N)
fine-tune with $\mathcal{L}_{\text{CE}}$
self-training: repeat for R rounds